Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	138	**1**	
1(a)(ii)	128	**1**	
1(b)(i)	135	**1**	
1(b)(ii)	121	**1**	
1(b)(iii)	134	**1**	
1(b)(iv)	125	**1**	
1(c)	24	**2**	**B1** for numerator of −24 or denominator of −1 or answer of –24
1(d)	$\frac{20\times\sqrt{9}}{30\div 6}$	**M1**	**M1** for all correct roundings
	12	**A1**	If 0 scored **SC1** for 3 correct roundings or 20.[0] and 9.0[0] and 30.[0] and 6.0[0]
2(a)	Trapezium	**1**	
2(b)	Enlargement [Scale factor] $\frac{1}{3}$ oe [Centre] (−5, −5)	**3**	**B1** for each
2(c)	$\frac{1}{9}$	**3**	**B2** for $\left(\frac{1}{3}\right)^2$ or **B1** for [shaded area] 13.5 or [area of A] 1.5 seen **M1** for $\frac{1.5}{their 13.5}$ oe
2(d)(i)	Image at (−6, 6),(−5, 6),(−5, 5), (−7, 5)	**2**	**B1** for image of A at $\begin{pmatrix}-4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\7\end{pmatrix}$
2(d)(ii)	Image at (1, 1), (1, 2), (3, 2), (2, 1)	**2**	**B1** for 180° rotation with incorrect centre

Question	Answer	Marks	Partial Marks
2(d)(iii)	Image at (5, −2), (5, −1), (6, −1), (7, −2)	**2**	**B1** for reflection in $y = 2$ or in $x = k$
3(a)	7 : 4 : 5	**2**	**B1** for any correct ratio other than 21 : 12 : 15, not in simplest form
3(b)(i)	$\frac{7}{16}$ oe $\times 1248$	**B1**	
3(b)(ii)	[Mustapha] 312 [Joshua] 390	**2**	**B1** for each or **M1** for $\frac{1248}{21+12+15} \times k$ where $k = 12$ or 15 or $\frac{1248}{their7 + their4 + their5} \times k$ oe where $k = their\ 4$ or $their\ 5$
3(c)	2912	**2**	**M1** for 1248 ÷ 3 [× 7]
3(d)	13 500 – 0.16 × 13 500 − 500 or 0.84 × 13 500 − 500	**M2**	**M1** for 0.16 × 13 500 or 0.84 × 13 500 seen
3(e)(i)	3 × 12 × 340	**B1**	
3(e)(ii)	12.9 or 12.91 to 12.92	**3**	**M2** for $\frac{12240-10840}{10840}[\times 100]$ or $\frac{12240}{10840} \times 100[-100]$ or $\left(\frac{12240}{10840} - 1\right)[\times 100]$ or **M1** for 12240 – 10840 or $\frac{12240}{10840}$ oe
4(a)	−6 4 4 0	**2**	**B1** for 2 or 3 correct
4(b)	Correct smooth curve	**4**	**B3FT** for 7 or 8 correct plots or **B2FT** for 5 or 6 correct plots or **B1FT** for 3 or 4 correct plots
4(c)	$x = 2.5$ cao	**1**	
4(d)(i)	−2 1 5.5	**2**	**B1** for 2 correct
4(d)(ii)	Correct continuous ruled line from $x = -1$ to $x = 6$	**2**	**B1FT** for 2 or 3 correct plots
4(d)(iii)	[x =] −0.6 to −0.4 and 3.9 to 4.1	**2**	**B1FT** for each